AutoTimes: Autoregressive Time Series Forecasters via Large Language Models

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Introduction

The Challenge

Undeveloped Time Series Foundation Models

- **Cause:** Limited availability of **Time Series Corpora** and underexplored Scalable Pre-training.

The Motivation

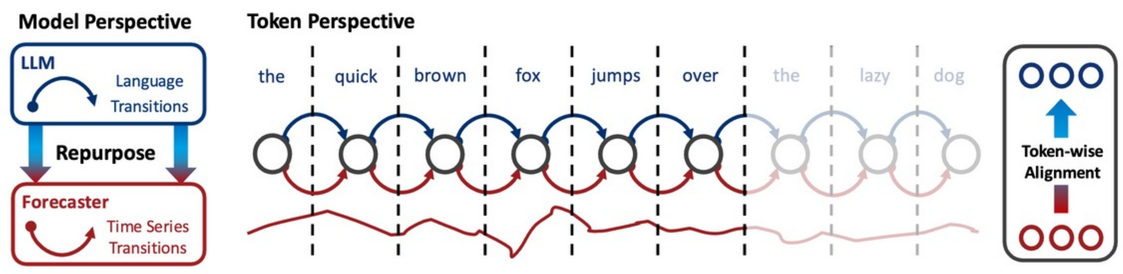
Revitalizing LLM Potentials

- **Gap:** Previous works neglected LLMs' inherent Autoregressive & Decoder-only nature.
- **Goal:** Leverage general-purpose **Token Transitions** for **Multi-step Generation**.

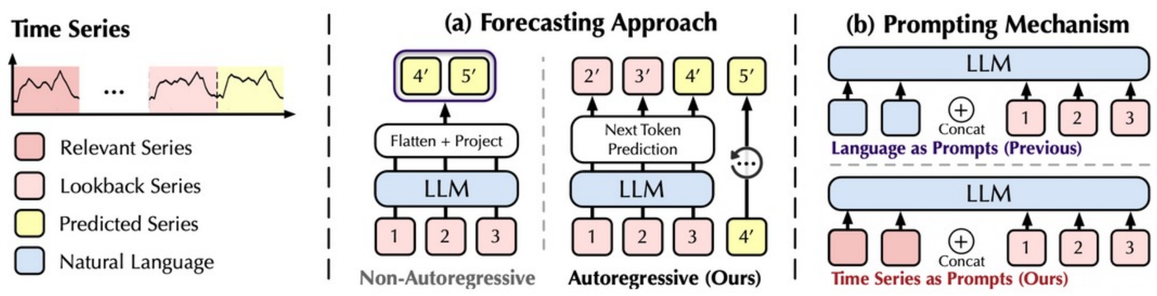
The Solution (AutoTimes)

Repurposing as Autoregressive Forecaster

- **Method:** Projects Time Series into Language Embedding Space.
- **Benefit:** Enables **Arbitrary-Length** prediction without case-by-case training.



We aims to truly repurpose Large Language Models as Large Time-Series Models that is different from previous non-autoregressive approaches:



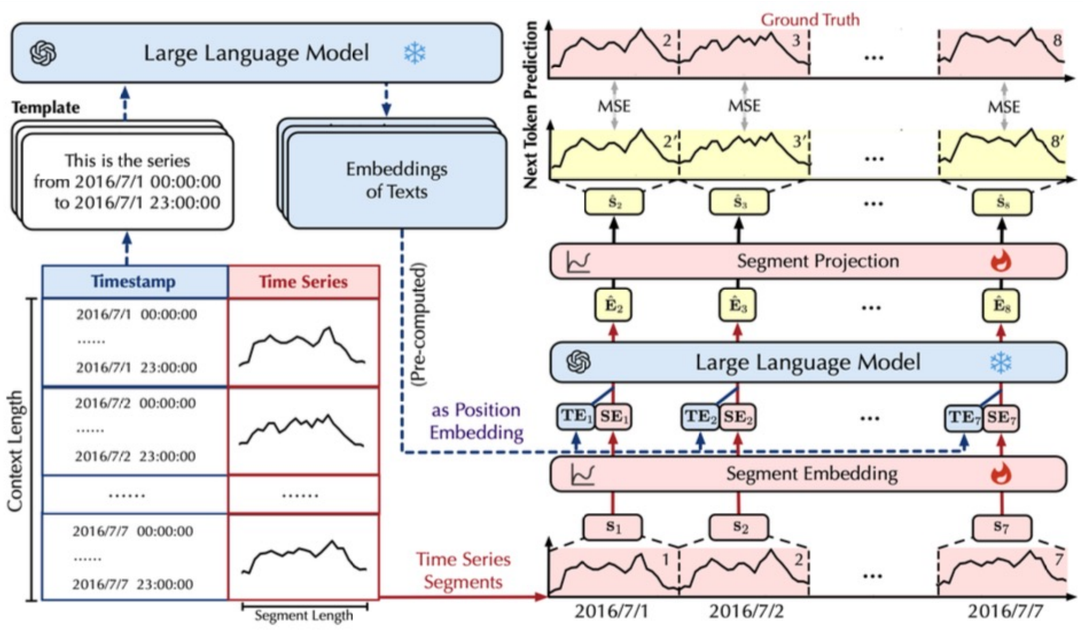
Architecture: Previous works adapt LLMs, which are generative decoders, as encoder-only forecasters with flattened tokens, breaking the causality.

Autoregression: LLM predicts the next tokens iteratively, while previous LLM-based forecasters predict all tokens in one single step, which apply only to specific input and output lengths, limiting model versatility.

Prompting: We retrieve time series as prompts, extending the context forprediction beyond the lookback series, termed in-context forecasting.

Multimodal: We use LLM-embedded meta-data (i.e., timestamps) toutilize chronological information and align multivariate time series.

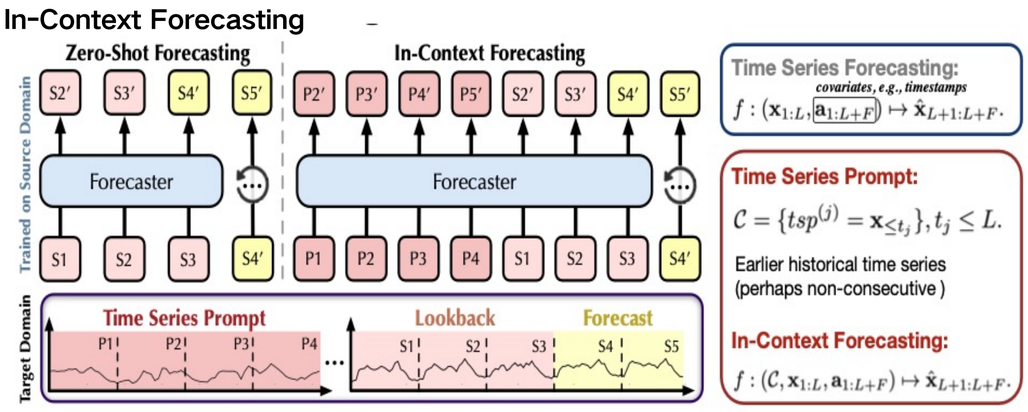
Overall Approach



- **Tokenization:** Regarding time series segments as basic tokens of the LLM1.
- **Position Embedding:** Embedded textual timestamps from language models.
- **Training Objective: Next Token Prediction**

- We adopt segment-level embedding to project time series tokens into the latent space: $\mathbf{SE_i} = \mathrm{SegmentEmbedding}(s_i)$ and add it with position embeddings: $\mathbf{E_i} = \mathbf{SE_i} + \mathbf{TE_i}$.
- We feed token embeddings $\mathbf{E_i}$ into the intermediate blocks of the frozen language model, getting next token representations: $\{\widehat{\mathbf{E_2}}, \ldots, \widehat{\mathbf{E_{N+1}}}\} = \mathrm{LLMLayers}(\{\mathbf{E_1}, \ldots, \mathbf{E_N}\})$.
- Segment-level projector converts representations to segments: $\hat{\mathbf{s}}_i = \mathrm{SegmentProjection}(\hat{\mathbf{E}}_i)$
- Finally, each predicted segment is supervised by the token-wise ground truth to optimize the parameters of time series embedding and projector: $\mathcal{L}_{\mathcal{MSE}} = \frac{1}{N_S}\sum |\mathbf{s_i} - \hat{\mathbf{s}}_i|_2^2$.

In-Context Forecasting



In-Context Learning: LLMs can generate better outputs based on **task demonstrations** from downstream datasets, without gradient updating.

- We realize this paradigm on time series forecasting by extending the context $\mathcal{C}$, which contains relevant **time series prompts**: $\mathcal{C} = \left\{ \mathrm{tsp}^{(j)} = \mathbf{x}_{\leq \mathbf{t_j}} \,\middle|\, \text{earlier historical time series} \right\}$.
- We concatenate time series prompts with lookback series and feed them as the input of the LLM-forecaster, termed in-context forecasting: $f: (\mathcal{C}, \mathbf{x_{1:L}}, \mathbf{a_{1:L+F}}) \mapsto \widehat{\mathbf{x}_{L+1:L+F}}$.

Main Experimental Results

Model	Task	Metrics
AutoTimes (based on GPT-2)	Long-Term & Short-Term Forecasting	MSE / MAE (Long-Term) SMAPE / MASE (Short-Term)

Long-Term Forecast Performance

Metric (MSE / MAE)	ETTh1	ECL	Exchange	Traffic	Weather	ILI
AutoTimes	**0.397 / 0.425**	**0.173 / 0.266**	**0.093 / 0.222**	**0.406 / 0.276**	**0.242 / 0.278**	**1.868 / 0.963**
PatchTST	0.413 / 0.431	0.159 / 0.253	0.092 / 0.221	0.391 / 0.264	0.226 / 0.264	1.956 / 0.995
DLinear	0.423 / 0.437	0.177 / 0.274	0.096 / 0.228	0.434 / 0.295	0.240 / 0.300	2.152 / 1.050

Conclusion: At the GPT-2 level, AutoTimes demonstrates SOTA or highly competitive performance on multiple benchmarks, including ETTh1, Exchange, and ILI.

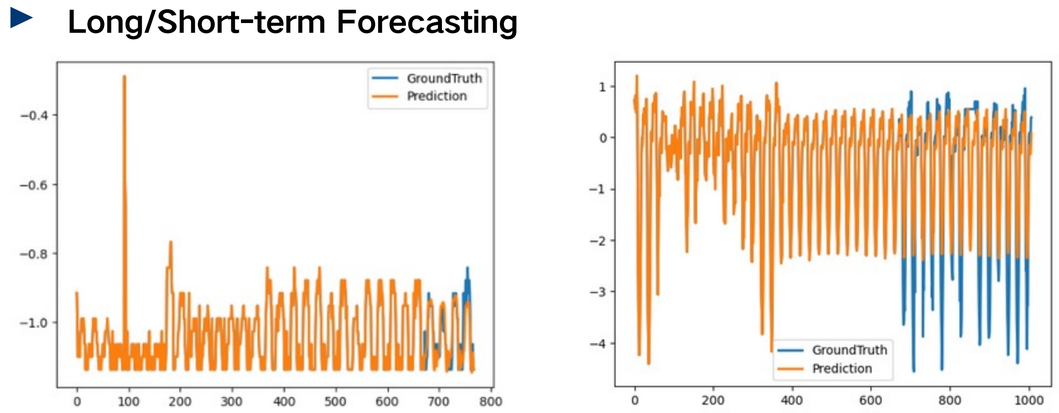
Short-Term Forecast Performance

Metric	Yearly	Quarterly	Monthly	Weekly	Daily	Hourly	All (Avg.)
SMAPE (↓)	13.3190	10.1010	12.7100	7.5190	3.3280	14.0090	**11.8310**
MASE (↓)	2.9930	1.1820	0.9340	0.7410	0.9700	2.7750	**1.5850**
OWA (↓)	0.7840	0.8900	0.8800	0.7630	0.9120	0.9610	**0.8500**

Conclusion: The model achieved an overall SMAPE of 11.8310 on the M4 dataset, proving especially effective on the Daily and Weekly subsets.

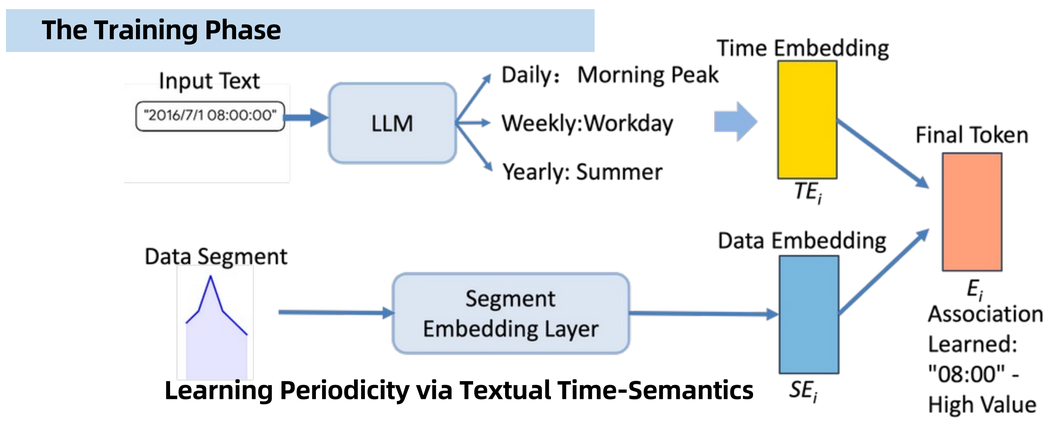
Forecasting Visualizations

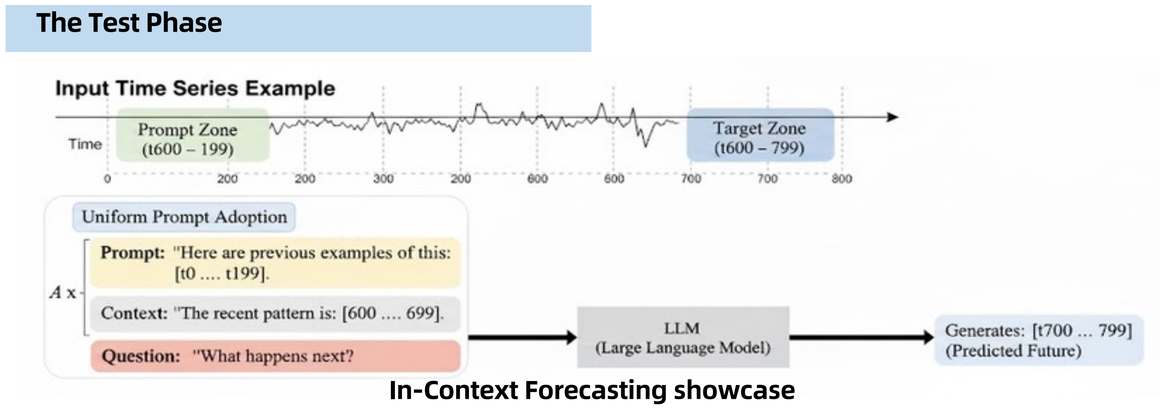
Long/Short-term Forecasting



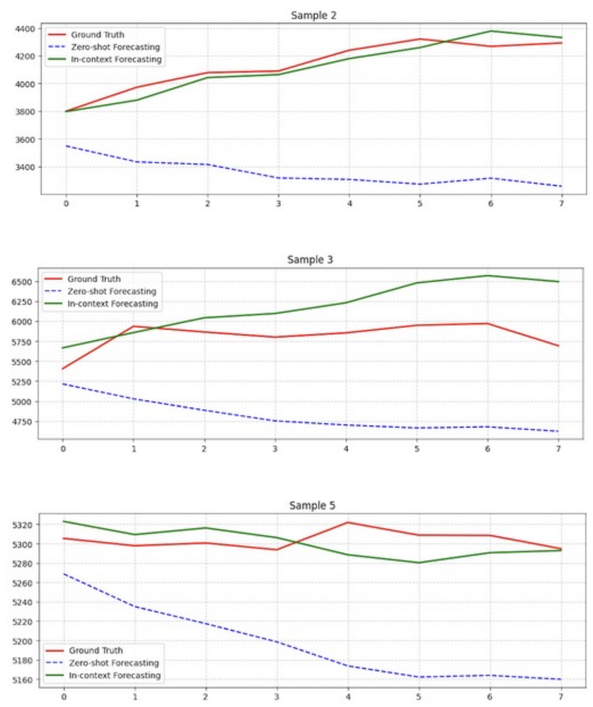
The Prediction and Ground Truth curves are closely aligned on both long-term (e.g., ETTh1, Traffic) and short-term (e.g., M4-Weekly) tasks. This visually demonstrates the model's versatility and high accuracy across different dataset types.

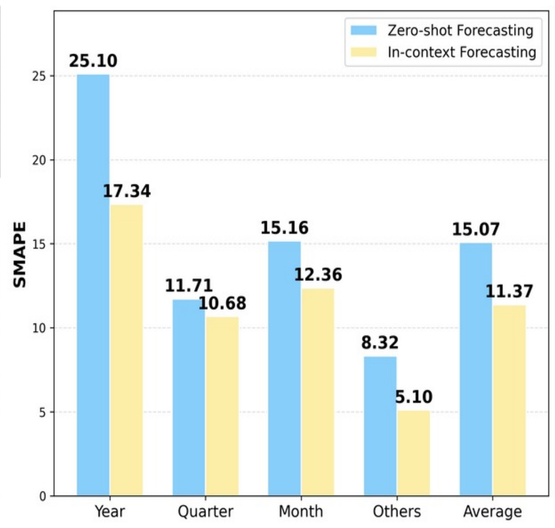
In-Context Forecasting





In-Context Forecasting showcase





Enhanced performance with prompts

Code Link: https://drive.google.com/drive/folders/1miW4KyAvJ1myZ9ymzV8uFRB-p50Mdn4k?usp=drive_link